Strengthening Accuracy of Lung Cancer Diagnosis Through A Novel Multimodal Bayesian Convolutional Neural Network

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2024 IEEE International Conference on Big Data (BigData) | 979-8-3503-6248-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/BigData62323.2024.10825351

***Abstract*—Lung cancer is a global epidemic, attributed to the general lack of early and effective screening, leading to low survival rates. Machine learning techniques have previously been used in diagnosing lung cancer, yet no comprehensive, integrated method of detecting lung cancer currently exists. This study proposes a novel method of lung cancer diagnosis through a multimodal process by combining both histopathological tissue images and CT scan images to diagnose lung cancer, while also utilizing bayesian layers within the neural network to account for prediction uncertainty. The proposed method achieved a test accuracy of 99.43%, performed better than traditional baseline methods, and was upheld as a robust method for strengthening current lung cancer diagnosis. The scope of the proposed method extends beyond just lung cancer, to other forms of cancer diagnosis and the entire biomedical imaging field.**

***Index Terms*—multimodality, lung cancer detection, bayesian convolutional neural network, neural network, lung cancer**

I. Introduction

Lung cancer is the most common cancer globally [1], and is the cancer with the highest mortality rate in the United States [2]. Unfortunately, more than 50% of lung cancer patients die within a year of diagnosis [3]. Survival rates are higher when detected early [4], but only 23% of lung cancer diagnoses are early stage diagnoses [5]. Early screening, especially through low-dose Computed Tomography (CT) scans, can reduce the risk of mortality by 20% [6].

Machine learning is key to more rapid and early detection of lung cancer [7], as it can be especially useful in factoring out variables that contribute to noise in the data, such as bones, tissue, and air in scans [8]. Classification algorithms, Linear Discriminant Analysis, and neural networks have all been utilized in Computer Aided Diagnosis (CAD) systems with CT scans to extract more information from nodules. Diagnosis can also be better visualized [7] through image processing with neural networks [8]. Due to the large genetic variety in lung cancer patients, machine learning has also been combined with diagnosis through genomic signatures for better diagnosis [7]. Overall, neural networks and deep learning seem to consistently promote the highest accuracies in lung cancer detection as compared to general machine learning algorithms [8].

Ultimately, there are many discrete solutions for lung cancer detection through various subfields of machine learning. However, there are no integrated systems for detection; that is, most current solutions utilize a single form of data for lung cancer diagnosis. It is thus key to combine multiple forms of imaging data, such as CT scans and histopathological data, to detect lung cancer [7]. Such a multimodal model would apply neural networks in a novel way to the field, and can strengthen diagnosis predictions by allowing for comprehensive review of multiple data sources, preventing physician misdiagnosis.

Moreover, taking into account the uncertainty of any algorithm is necessary, especially in a medical diagnosis scenario. Adding a bayesian component to a multimodal neural network would only strengthen a lung cancer diagnosis model's training and promote caution in medical applications.

Thus, this study aims to utilize neural networks to detect lung cancer in patients by creating a multimodal model combined with bayesian inference that utilizes both CT scans of the lung and histopathological tissue images to diagnose adenocarcinoma, squamous cell carcinoma, and benign tumors. It is hypothesized that the multimodal model will yield greater test accuracies as compared to the test accuracies of individual neural network models for detecting lung cancer using either CT scans or histopathological images.

II. Methods

Both CT scan data and histopathological tissue image data was required to train the multimodal model. For the CT scan data, data from Hany 2020 [9] and Al-Yasriy et al. 2023 [10] were compiled together. For the histopathological tissue images, data from Borkowski et al. 2019 [11] was used. The images were sized to [224, 224] and rescaled. The shear and zoom ranges were set to 0.2, and horizontal flip transformations of the image were performed, allowing the neural networks to be exposed to variations of the data. The images consisted of adenocarcinoma images, squamous cell carcinoma images, and benign tumor images; these served as the main classes for neural network classifications.

To compare the multimodal model to current traditional neural networks, baseline Convolutional Neural Network (CNN) models were built. The first CNN model was built solely trained on CT scan images. The architecture of this CNN consisted of five convolutional layers, four pooling layers, and 2 fully connected layers. The CT scan model was trained

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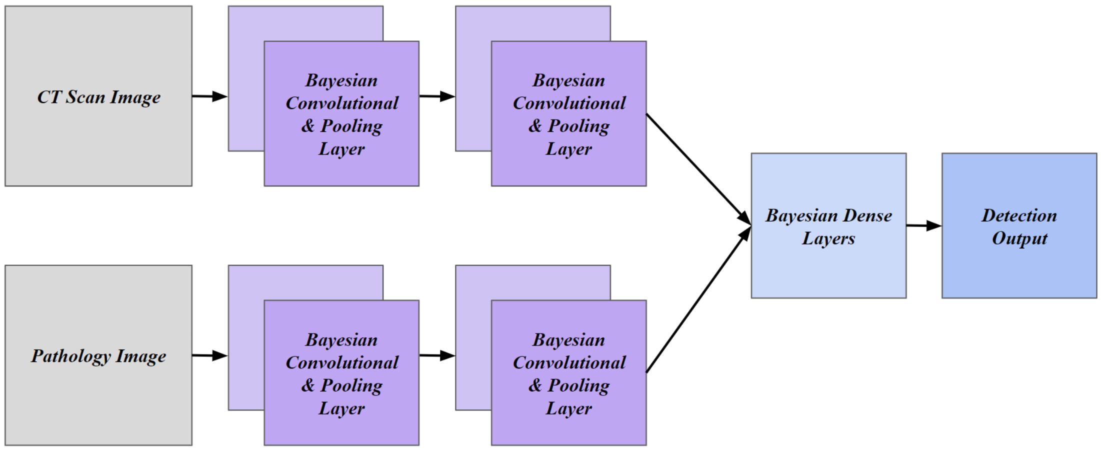

Fig. 1. The multimodal neural network architecture built in this study is represented in a visual manner. Each branch contains two bayesian convolutional layers and two pooling layers, where the CT scans travel through one branch and the histopathology images travel through the other. The branches are then concatenated and two bayesian dense layers are incorporated, before the model delivers a lung cancer diagnosis prediction.

with an adaptive moment estimation optimizer, Adam, and a Categorical Cross Entropy Loss function for 50 epochs with only the CT scan training images. Testing was conducted through just the CT scan test images. The same was repeated for the baseline histopathology CNN. The same layers, hyperparameters, and number of training epochs were used. The histopathology model was trained with only the histopathology training images, and testing was done through the histopathology test images.

The multimodal model received both the CT scan and the histopathology training images as input. To process both types of images, two different pathways for the neural network were built: one for the CT scan images and the other for the histopathology images, as can be seen in Fig. 1 by the two separate branches. Bayesian convolutional layers were constructed and incorporated to account for the uncertainty that comes with medical image diagnosis, limit overfitting, and promote data efficiency. Each pathway consisted of two bayesian convolutional layers and two pooling layers, so that both types of images were processed under similar network constraints. The two pathways were then concatenated together, and two bayesian dense layers were added. The same hyperparameters as the baseline model were used: an Adam optimizer and Categorical Cross Entropy Loss function. The multimodal model was trained for 50 epochs, and then tested on both the CT scan and histopathology test images.

For each model, the final training accuracy and final training loss were recorded, as well as the test loss. The main metric for measuring the performance of the models was the test accuracy, as it indicates how well the model performs on unseen data.

III. Results and Discussion

When considering the training accuracies, as seen in Table 1, the multimodal model achieved the highest training accuracy, yet the range of the training accuracies across the models differed by less than 1%. The multimodal model also had the lowest training loss, upholding the idea that both the multimodal and bayesian aspect of the neural network can improve lung cancer diagnosis accuracy. Regardless, all three training accuracies were high and the training losses were low. When looking at the training graphs, as seen in Fig. 2, it appears that there are slightly fewer fluctuations with the multimodal model than the baseline models.

The main metric for performance in this study was the test accuracy, as it is a true measure of the model's accuracy in diagnosing lung cancer on unseen data. The histopathology model achieved a test accuracy that was lower than the other two models, which is reflected in the high test loss of the histopathology neural network model. This presents the possibility that the histopathology model may have overfit the training data. The multimodal model achieved the highest test accuracy of 99.43% the three models, and the lowest test loss of 0.0164. The results from this study support the hypothesis that the multimodal model can strengthen the accuracy of classical neural network methods of diagnosing lung cancer. The advantage of the multimodal model is the concatenation of the two different branches and receiving prediction probabilities from both types of data in order to create stronger predictions, which allows for a validation system should one branch fail in accurately classifying lung cancer for a given scenario. The bayesian aspect, which takes into account uncertainties in the data and promotes data efficiency, also contributed to the high test accuracy.

While the multimodal model achieved greater test accuracies than the baseline models in this study, it is also necessary to compare the model to other recent studies exploring neural network methods for detecting lung cancer. As shown in Table 2, the multimodal model is continuously upheld as a strong option with a higher or similar accuracy to other existing models of lung cancer detection.

As Li et al. [7] emphasize the importance of integrating

TABLE I
LOSS AND ACCURACY OF MODELS

	CT Scan Model	Pathology Model	Multimodal Model
Training Loss	0.0458	0.0392	**0.0229**
Training Accuracy	98.28%	98.69%	**99.10%**
Test Loss	0.0377	0.2702	**0.0164**
Test Accuracy	97.70%	93.68%	**99.43%**

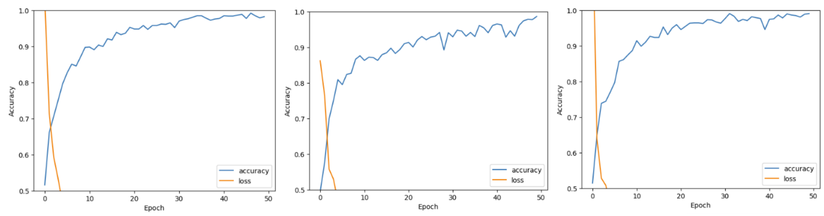

Fig. 2. Training graphs of the CT Scan, histopathology, and multimodal models, Respectively across the 50 training epochs, depicting both the training loss and training accuracy progression.

TABLE II
COMPARISON TO SIMILAR NEURAL NETWORK STUDIES

Authors	Summary	Accuracy
Khattar et al. 2024 [12]	Nodule detection in CT scans through traditional CNN	95.80%
Rehman et al. 2023 [13]	CAD of lung tumors in CT scans through CNN	98.33%
Atiya et al. 2023 [14]	Transfer learning method ofthrough CT scans	96.12%
Wahid et al. 2023 [15]	CAD of lung cancer in pathological images through CNN	98.82%
Faria et al. 2023 [16]	Novel CNN diagnosis through pathological images	88.50%
Wani et al. 2024 [17]	Interpretable convolutional and XGBoost neural network	97.43%
Reddy et al. 2023 [18]	Improved dial's loading algorithm CNN through CT scans	92.81%
This study	**Multimodal bayesian convolutional neural network model**	**99.43%**

data types in order to overcome current shortcomings where traditional neural network models overlook the comprehensive amounts of data used in making clinical decisions, the multimodal method proposed holds as a strong approach for addressing this issue.

Despite this, it is necessary to address a few limitations in the scope of the experiments conducted in this study. The main limitation in this study was the lack of availability of data, especially publicly available CT scan datasets. The limited availability of CT scans led to the concatenation of two CT scan datasets for the training and testing datasets, and also limited the number of histopathology images that could be used, as the same number of both types are necessary to train the multimodal model.

Furthermore, there are currently no public datasets that provide both the CT scan and the histopathological image of the same patient. Further testing should be conducted to uphold the findings of this study when data availability increases. Moreover, due to the large number of epochs that the models were trained on, they could have been subject to some overfitting. However, the bayesian layers in the multimodal model likely reduced the effects of overfitting by modeling uncertainty in the weights of the neural network. Additional data availability could also help solve this issue, while also exposing the models to wider ranges of lung cancer data and other types of lung cancer.

Also, due to the high computational complexity and cost of training the bayesian convolutional neural network, modifying and improving upon the current model would involve significant resources. Simplifying the architecture should be a next step to reducing computational complexity, which could be necessary if this model were to be implemented in a clinical setting.

IV. CONCLUSION

This study initially proposed a multimodal bayesian method of detecting lung cancer through the use of both CT scan images and histopathological tissue images. When compared to baseline neural network models, the hypothesis that the multimodal model would achieve a higher accuracy was supported. As such, the proposed architecture holds great promise within lung cancer diagnosis.

Multimodality can serve as a method of validating current physician diagnosis of lung cancer. As physicians often use multiple sources of information for diagnosis, the multimodal model offers a comprehensive opportunity to allow for CAD systems within the healthcare field. Combined with the consideration of bayesian inferences and probabilities, such a neural network model could be key in increasing early detection of lung cancer and increasing survival rates and prognosis [6].

As with any technology that assists with medical decisions, including the proposed model, a physician perspective is necessary. Especially within diagnosis, the proposed model is not 100% accurate, and there is always a change of false diagnoses. While machine learning and neural network tools such as the multimodal model can be used to aid lung cancer detection and biomedical image diagnosis, the primary diagnosis should come from a human physician.

The neural network can be further expanded by training and testing on other types of lung cancer, such as large cell carcinoma, and also new network architectures, such as using separable convolutional layers. The concept can also be applied to other fields of cancer and general biomedical image processing. Further expansions of the multimodal nature of the proposed method could involve the segmentation of the histopathological images to identify cellular and tumor regions, and the application of this segmentation to CT scan images to present information to patients in a more digestible manner. Incorporating Explainable AI into the model, such as Shapley Additive Explanations (SHAP) or Local Interpretable Model-agnostic Explanations (LIME) would further explain the decision making process of the model, which is key when implemented into a healthcare setting.

Overall, not only does the proposed multimodal neural network method hold promise in strengthening accuracy not only within lung cancer, but also in a variety of fields, and it can be used in novel ways to improve biomedical image diagnosis, segmentation, and prognosis.

Acknowledgment

I thank my mentor, Dr. Sonali Jindal, a cancer pathologist at the MD Anderson Cancer Center, for providing feedback and valuable insight throughout the research process. I would like to acknowledge Borkowski et al. 2019 [11] for the histopathological lung cancer images, and Hany 2020 [9] and Al-Yasriy et al. 2023 [10] for the CT scan images. I thank all the patients whose images have been used for this research work and for their contribution to accelerate our understanding of lung cancer. The author declares no competing interests or sources of funding for the work in this study.

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